Using spaCy for Natural Language Processing and Visualisation

spaCy is an open source Python library that lets you break down textual data into machine friendly tokens. It has a wide array of tools that can be used for cleaning, processing and visualising text, which helps in natural language processing.

Natural language processing (NLP) is an important precursor to machine learning (ML) when textual data is involved. Textual data is largely unstructured and requires cleaning, processing and tokenising before a machine learning model can be applied to it. Python has a variety of NLP libraries available for free such as NLTK, TextBlob and Gensim. However, spaCy stands out when it comes to its speed of processing text and applying beautiful visualisations to aid in understanding the structure of text. spaCy is written in Cython; hence the upper case 'C' in its name.

This article discusses some key features of the spaCy library in action. By the end of this article, you will have hands-on experience in using spaCy to apply all the basic level text processing techniques in NLP. If you are looking for a lucrative career in machine learning and NLP, I would highly recommend adding spaCy to your roster.

Installing spaCy

As with most of the machine learning libraries in Python, installing spaCy requires a simple *pip* install command. While it is recommended that you use a virtual environment when experimenting with a new library, for the sake of simplicity, I am going to install spaCy without a virtual environment. To install it, open a terminal and execute the following code:

```
pip install spacy
python -m spacy download en_core_web_sm
```

spaCy has different models that you can use. For English, the standard model is *en_core_web_sm*. For the coding part, you will have to select a Python IDE. I personally like to use Jupyter Notebook, but you can use any IDE that you are comfortable with. Even the built-in Python IDLE works fine for the code covered in this article.

Once you are on your IDE, test if the installation has been successful by running the following code. If you get no errors, then you are good to go.

```
import spacy
myspacy = spacy.load('en_core_web_sm')
```

The *myspacy* object that we have created is an instance of the language model *en_core_web_sm*. We will use this instance throughout the article for performing NLP on text.

Reading and tokenising text

Let's start off with the basics. First create some sample text and then convert it into an object that can be understood by spaCy. We will then apply tokenisation to the text. Tokenisation is an essential characteristic of NLP as it helps us in breaking down a piece of text into separate units. This is very important